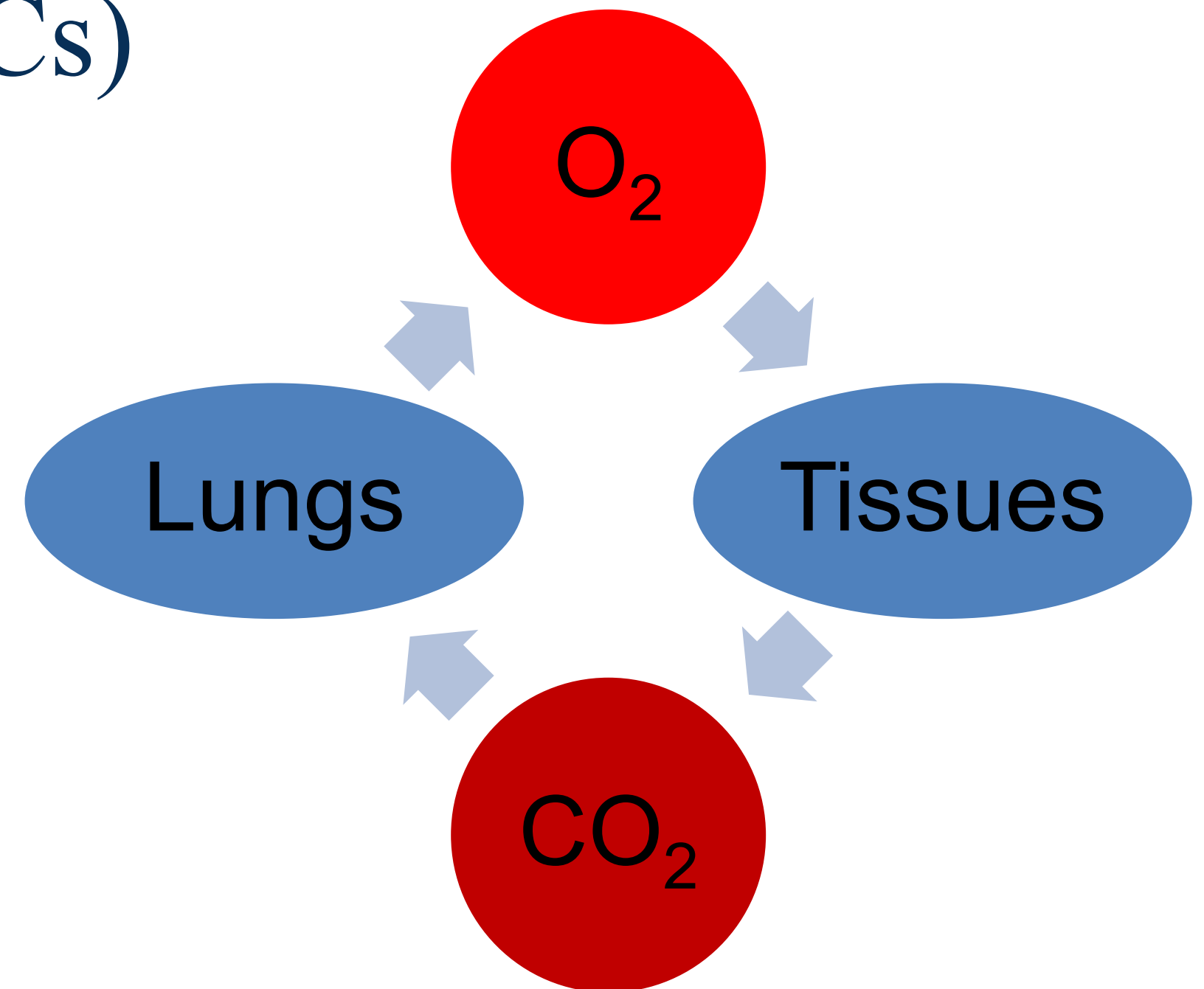


Erythrocyte

Erythrocytes (RBCs)

- Carries O_2 from lungs to tissues
- Carries CO_2 from tissues to lungs
- Anemia – insufficient number of RBCs
 - Tissue hypoxia – inadequate tissue oxygenation
- Erythrocytosis – excess number of RBCs
 - No adverse effect on pulmonary gas exchange

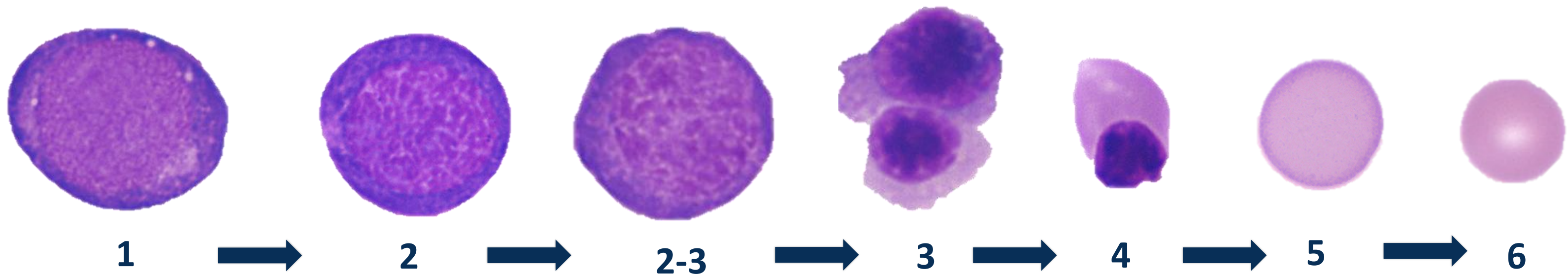


Life Cycle

- Immature
 - EPO is major cytokine regulating erythropoiesis
- Mature
 - Circulating life span of mature RBCs ~ 100–120 days
- Old
 - Senescent cells are destroyed by reticuloendothelial system macrophages
 - Occurs in spleen, liver, bone marrow

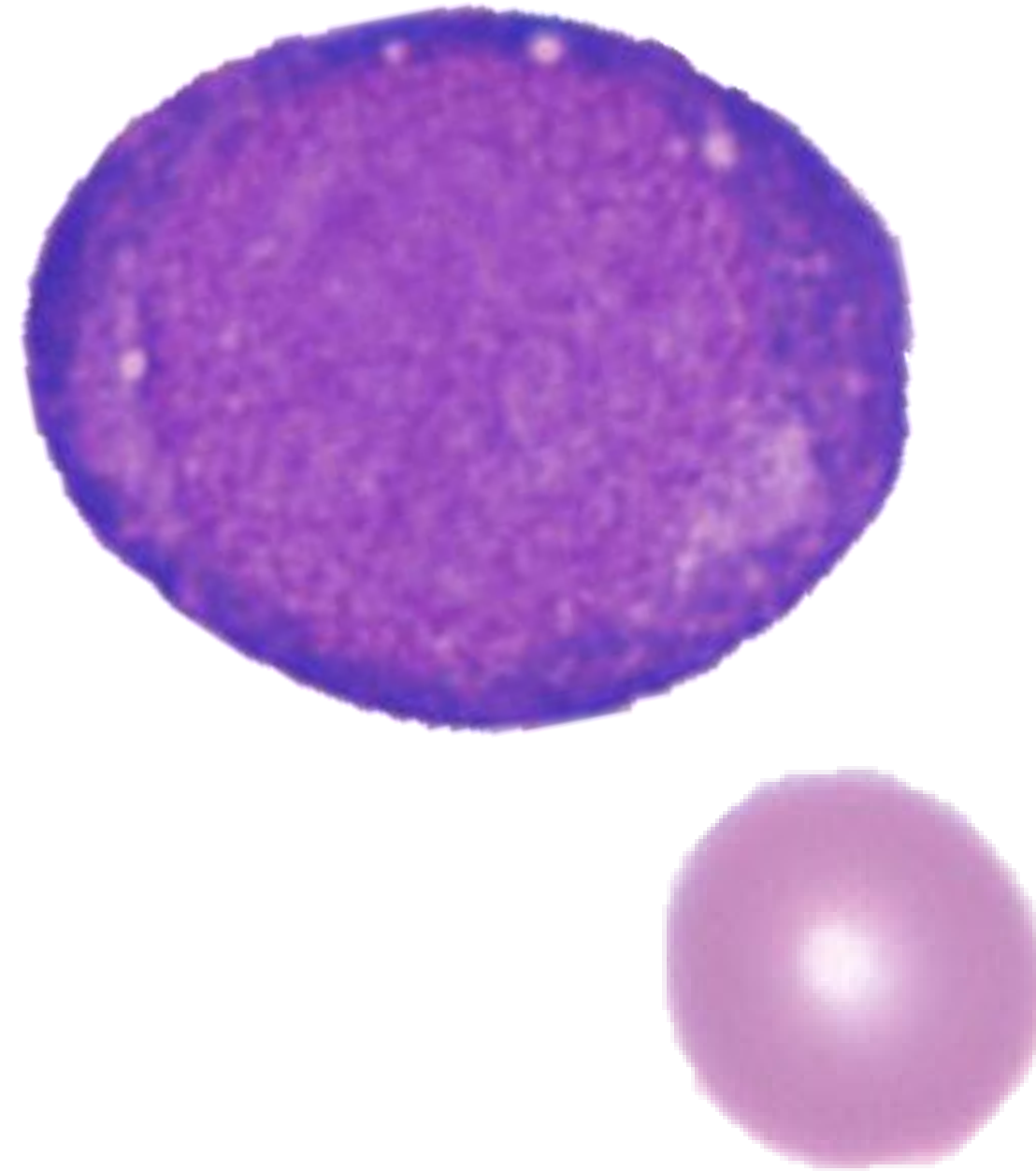
RBC Maturation

1. Gradual decrease in cell size with progressive condensation of the nuclear chromatin
2. Cytoplasm in younger cells is deeply basophilic due to the abundance of ribosomes
3. Increase in hemoglobin (acidophilic) as the cell matures, cytoplasm appears pink or salmon color
4. Eventual expulsion of the pyknotic nucleus



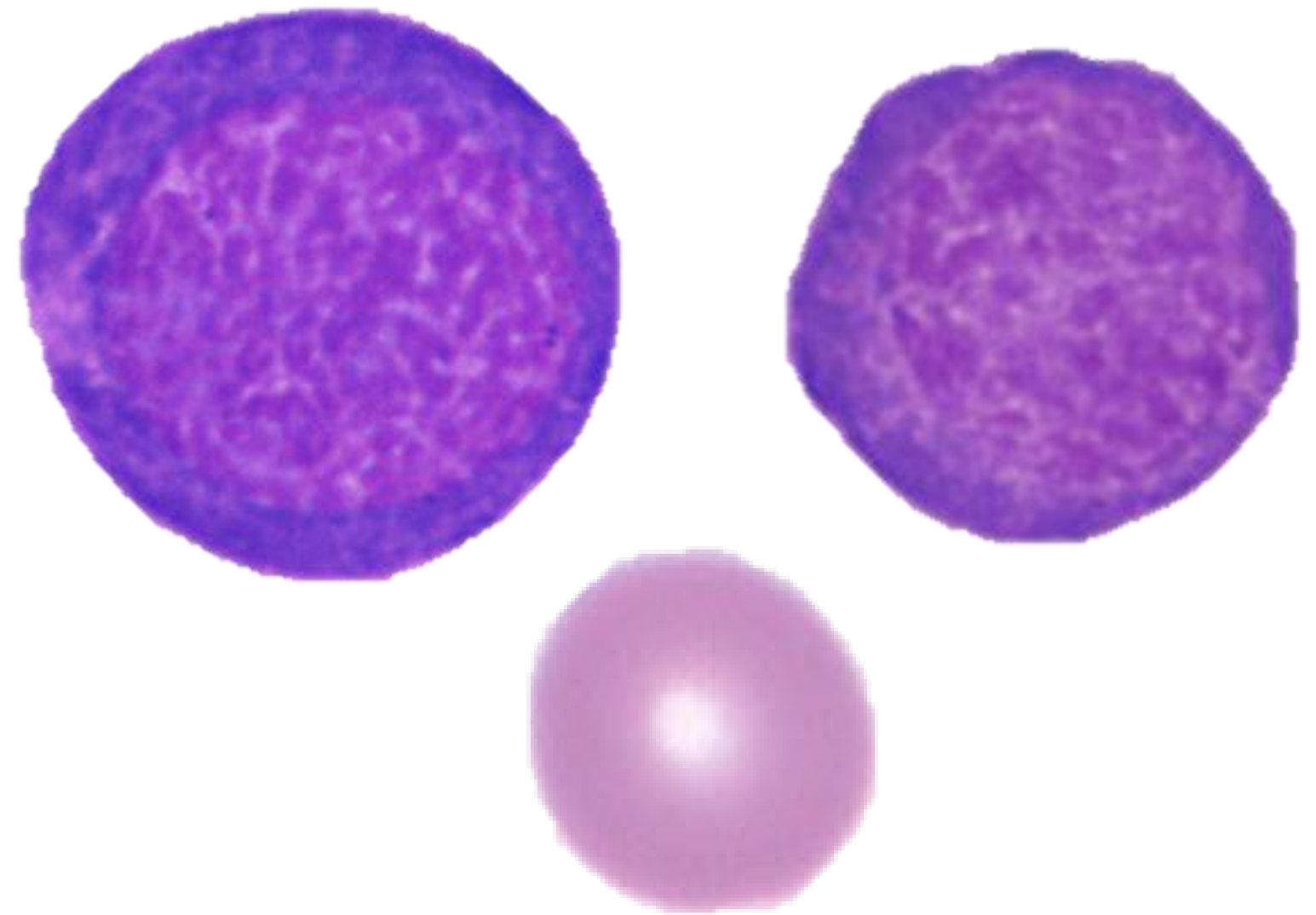
Pronormoblast (Rubriblast)

- ↑↑↑ N:C ratio
- Cytoplasm
 - ↑↑↑ ribosomes (deep basophilia)
 - Golgi apparatus
- Nucleus
 - Fine chromatin (lacy)
 - 1 – 3 nucleoli



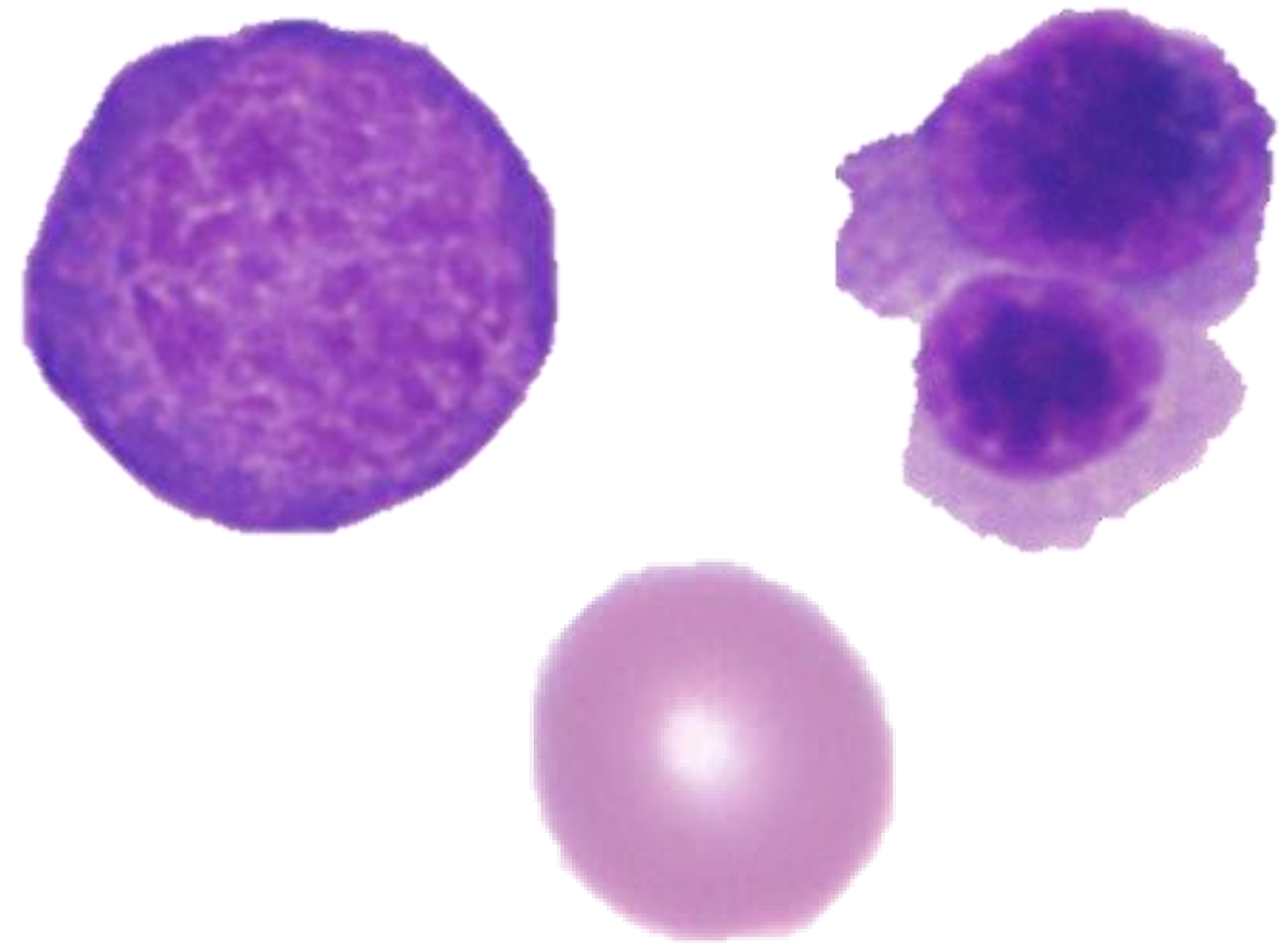
Basophilic normoblast (Prorubricyte)

- ↑↑ N:C ratio
- Cytoplasm
 - ↑↑ ribosomes (deep basophilia)
 - Lighter with maturity
 - Perinuclear halo (mitochondria)
- Nucleus
 - Coarse/fine chromatin (wheel spoke)
 - Nucleoli not obvious



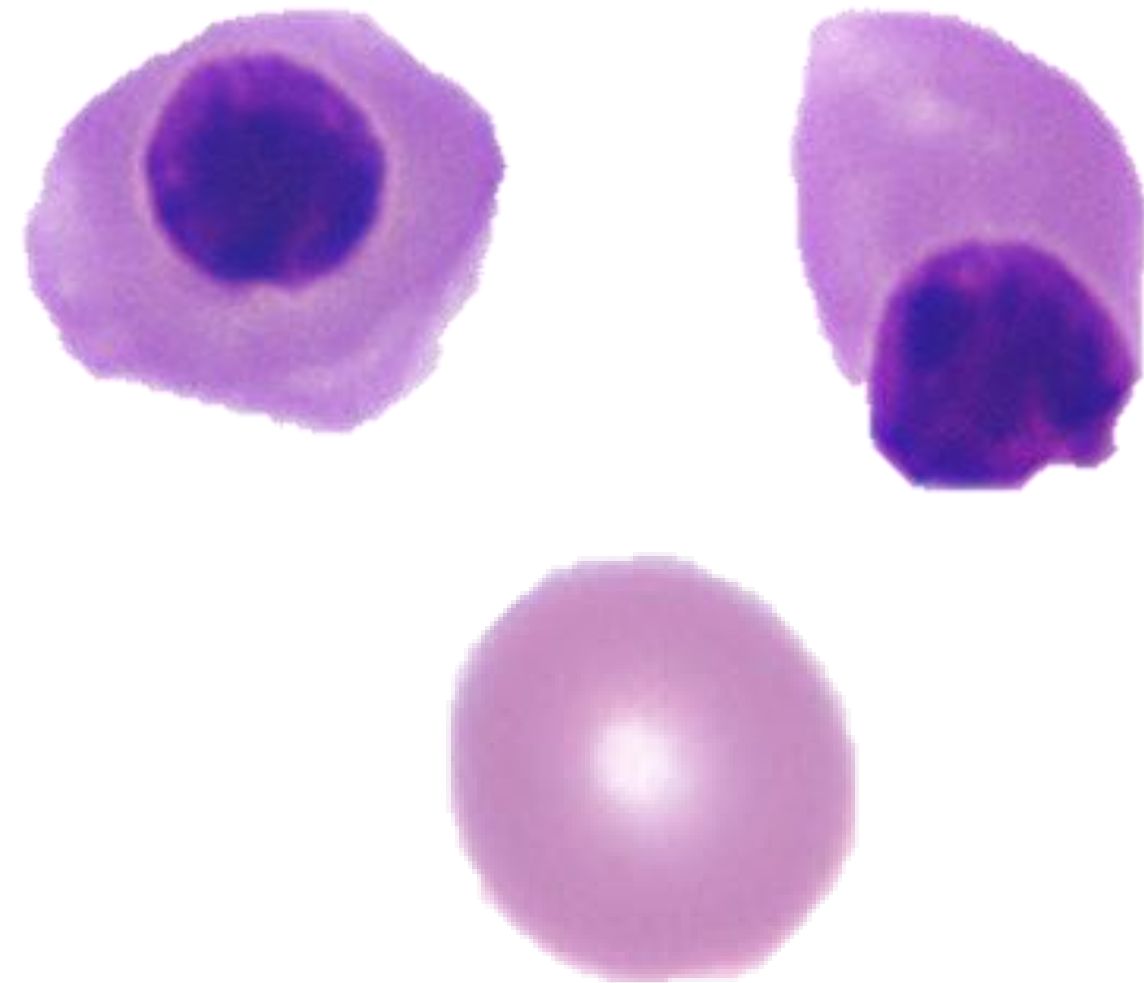
Polychromatic normoblast (Rubricyte)

- Decreasing N:C ratio
- Cytoplasm
 - Fewer ribosomes (gray-blue)
 - ↑ hemoglobin
- Nucleus
 - Chromatin condensation
- Final mitotic stage



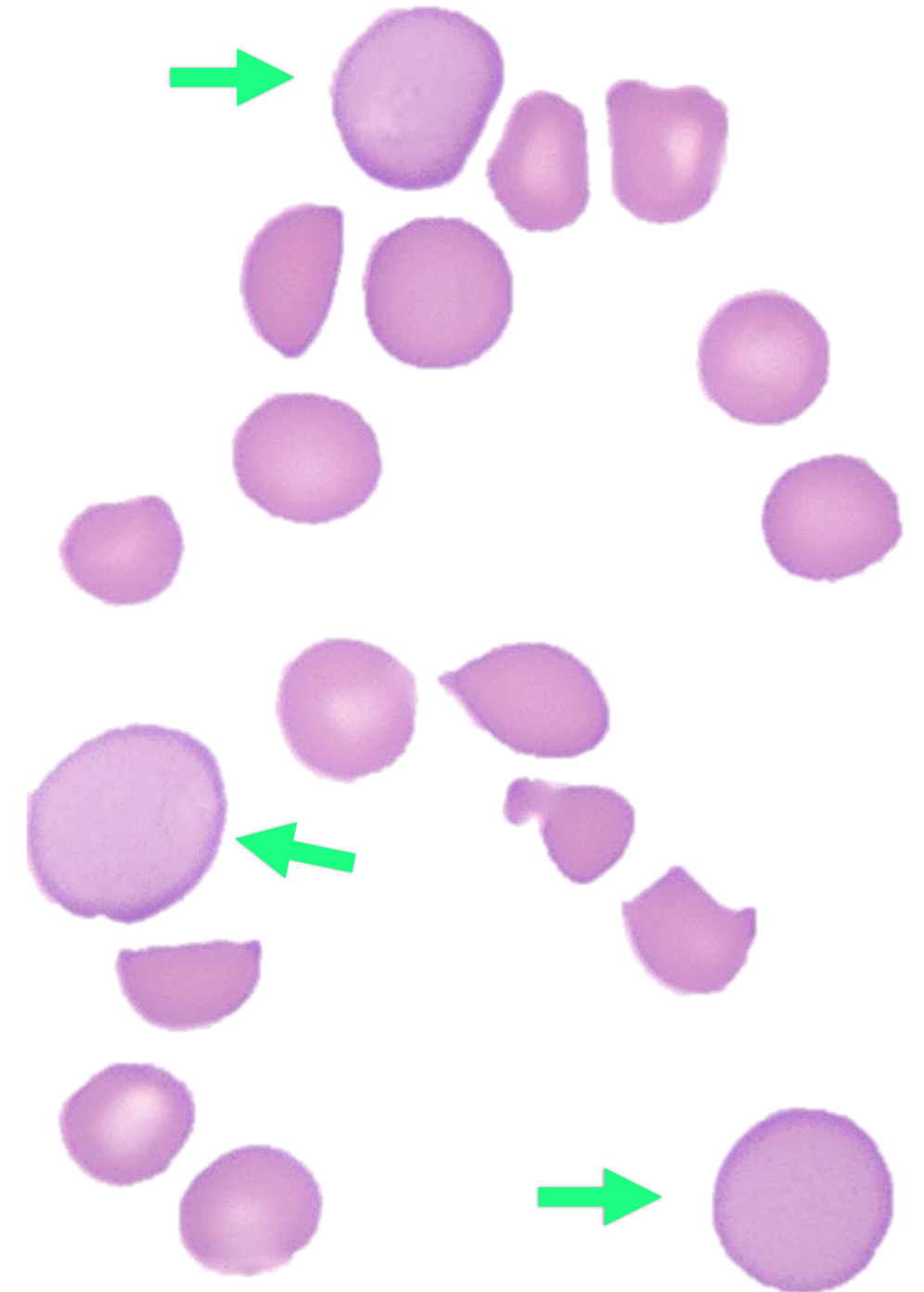
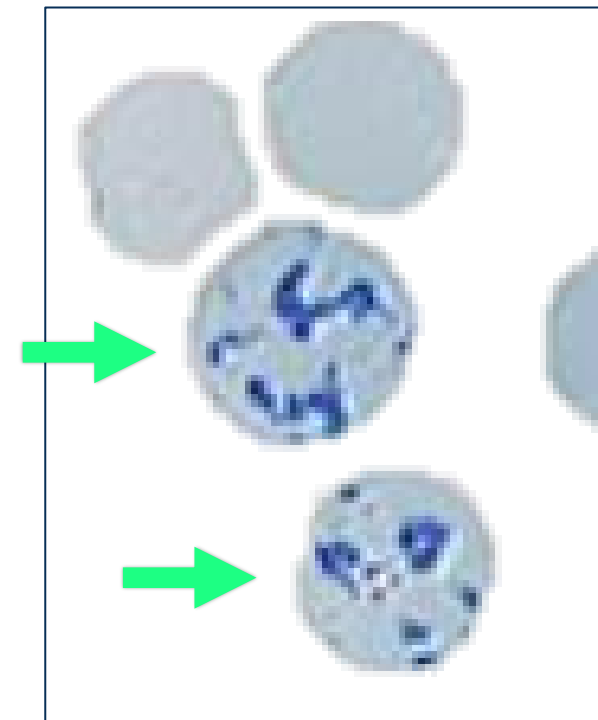
Orthochromic normoblast (Metarubricyte)

- ↓ N:C ratio
- Cytoplasm
 - ↑↑ hemoglobin (pink/salmon color)
- Nucleus
 - Chromatin heavily condensed
 - Pyknotic
 - Eccentric or partially extruded



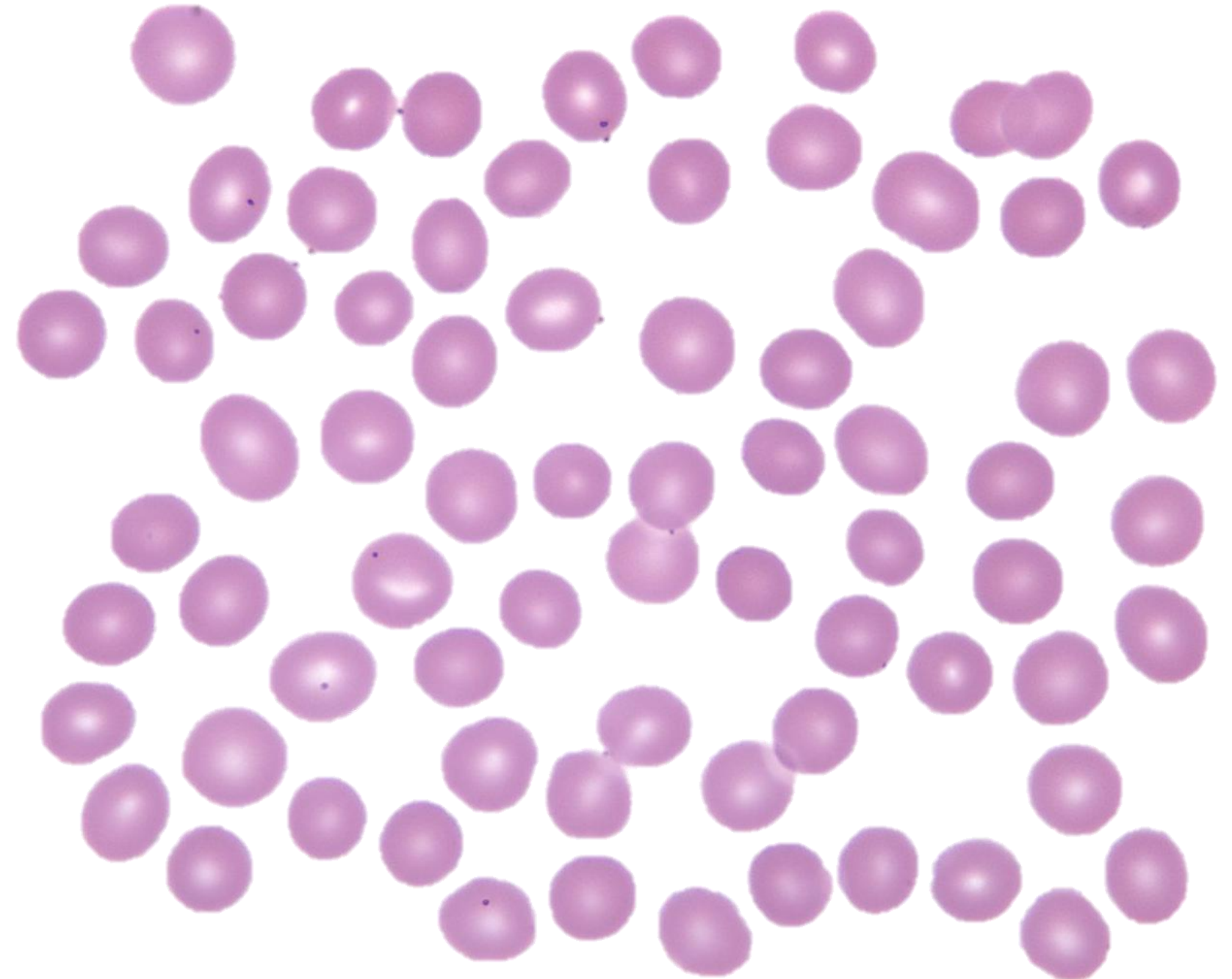
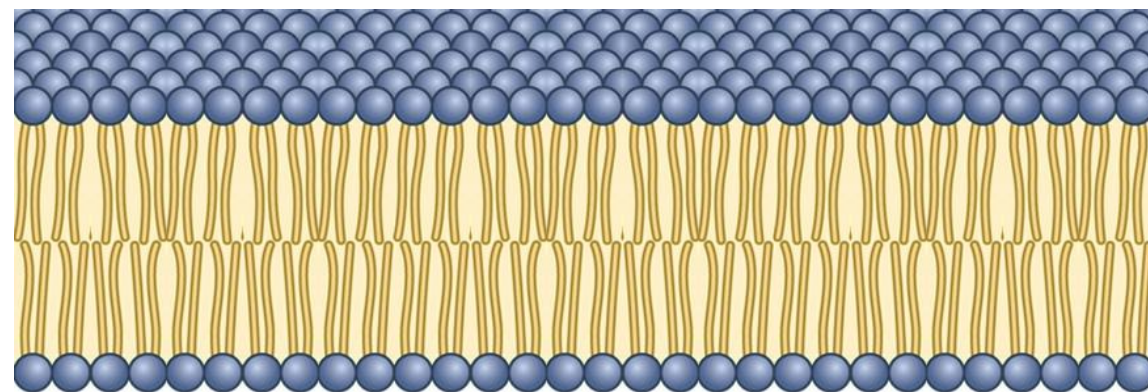
Reticulocyte

- Post extrusion, membrane remodels
- Cytoplasm
 - Bluish tinge
 - Polychromasia evaluation
- Supravital stains
 - Precipitates RNA and mitochondria
 - New methylene blue
 - Brilliant cresyl blue



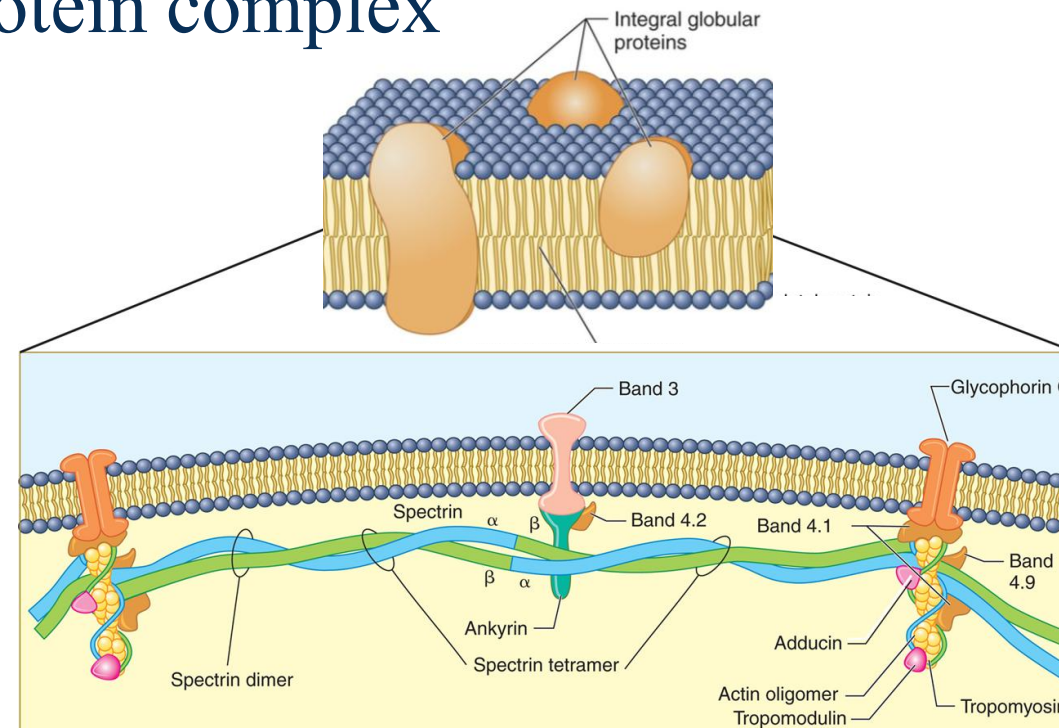
Erythrocyte

- Biconcave disc
- Lack organelles to synthesize new lipids and proteins
 - Membrane damaged RBCs culled or pitted by spleen

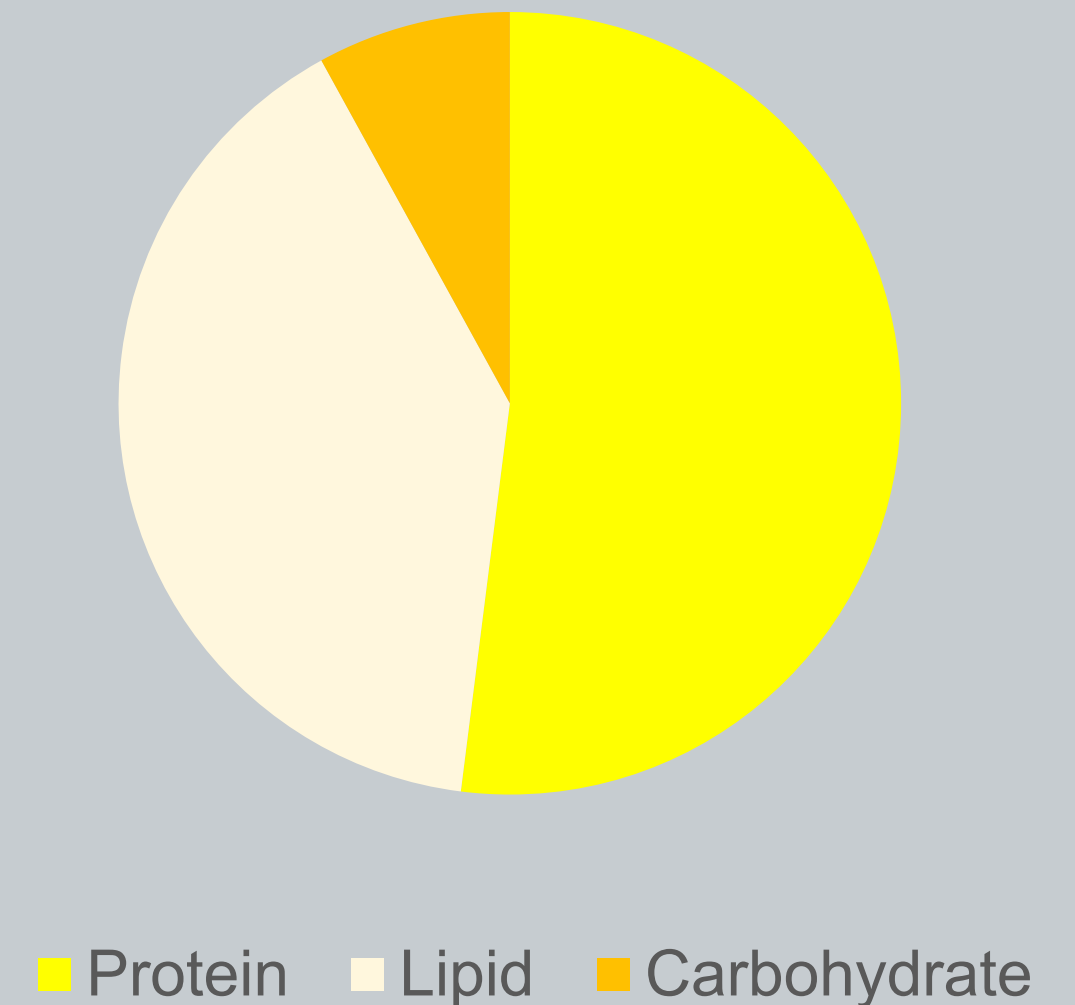


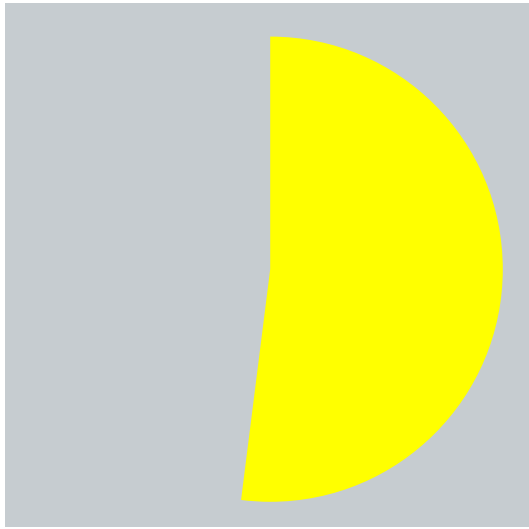
Erythrocyte Membrane

- Balances exchange of bicarbonate and chloride ions
 - Controlled permeability maintains osmotic equilibrium
- Cytoskeleton provides strength and flexibility
- Phospholipid bilayer-protein complex
 - ~ 52% protein
 - ~ 40% lipid
 - ~ 8% carbohydrate

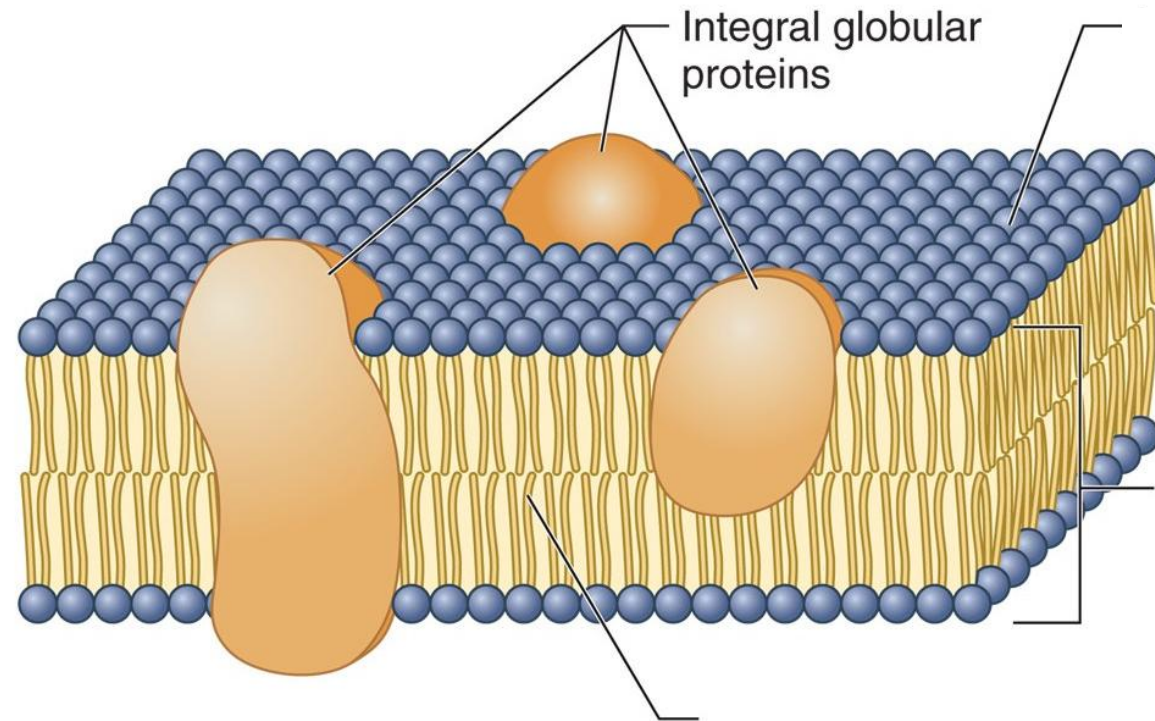


Phospholipid Composition





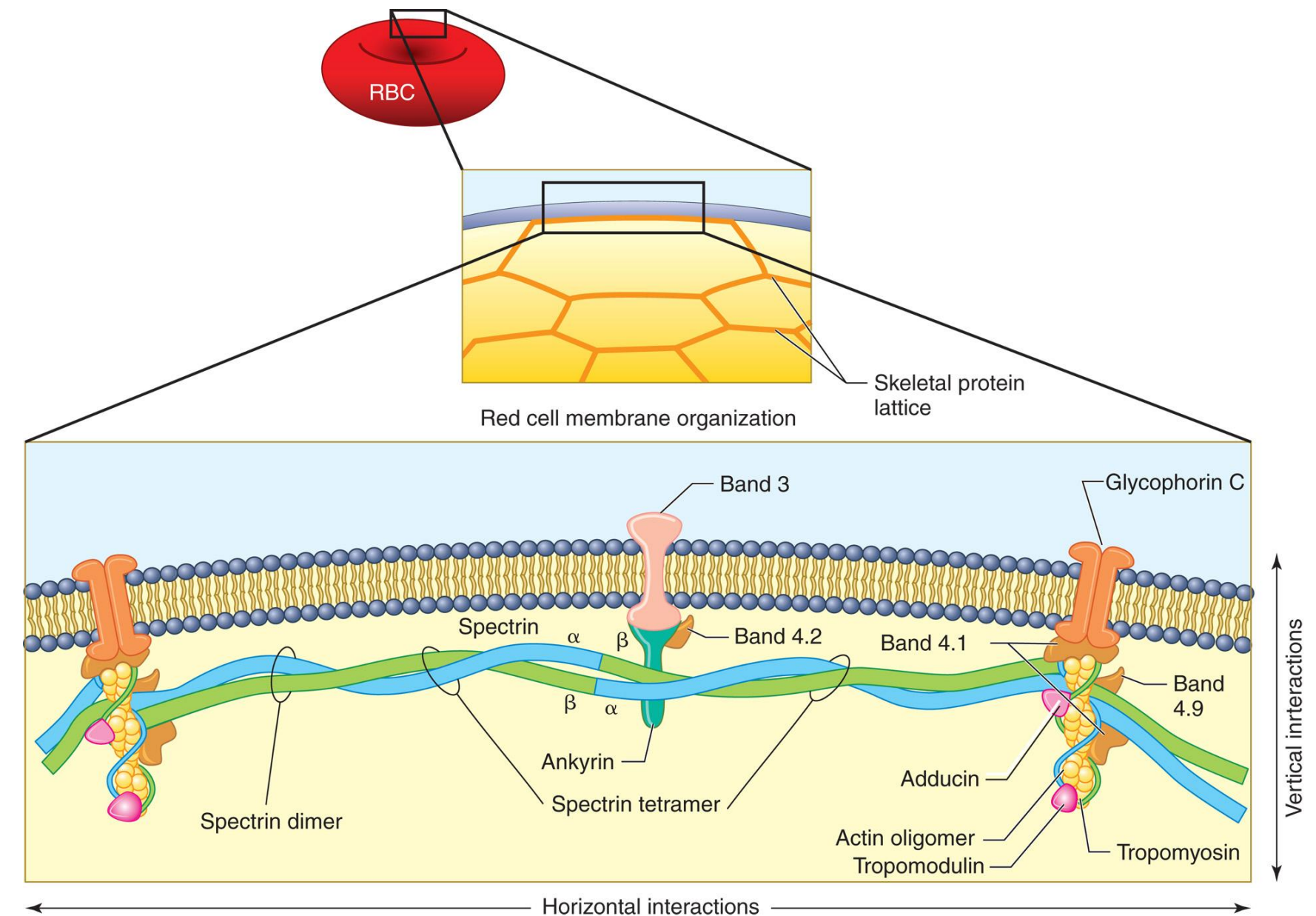
Protein Composition

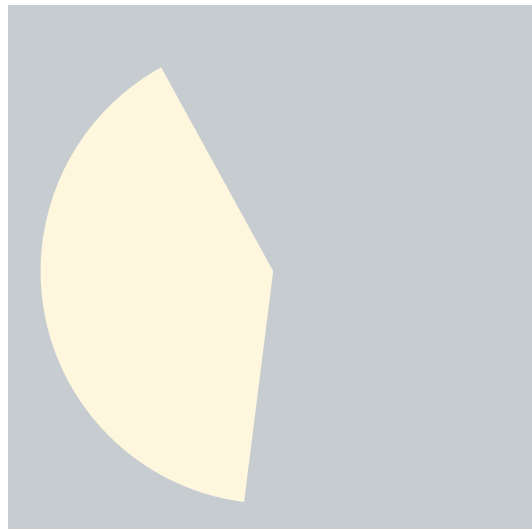


- Integral proteins
 - Extracellular domain (exterior surface)
 - Responsible for zeta potential (negative surface)
 - Prevent RBCs sticking
 - Band 3
 - Major binding site for cytoplasmic membrane components
 - Transport channel for chloride-bicarbonate exchange

Protein Composition

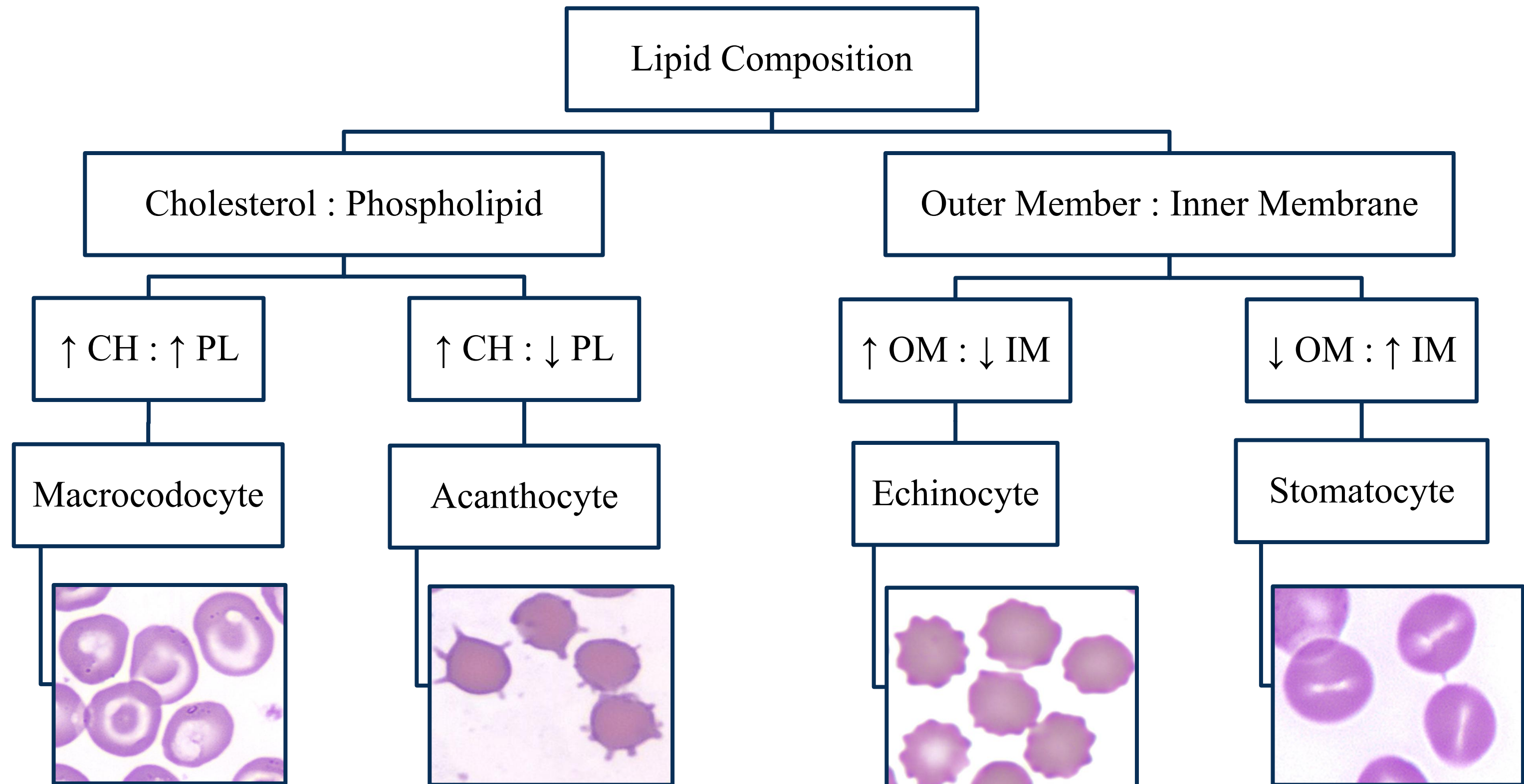
- Peripheral proteins
 - Includes enzymes and structural proteins (SPs)
 - Spectrin (SP)
 - Predominant skeletal protein
 - Functions like a spring
 - Ankyrin (SP)
 - Binds to Band 3
 - High-affinity binding site of spectrin to inner surface



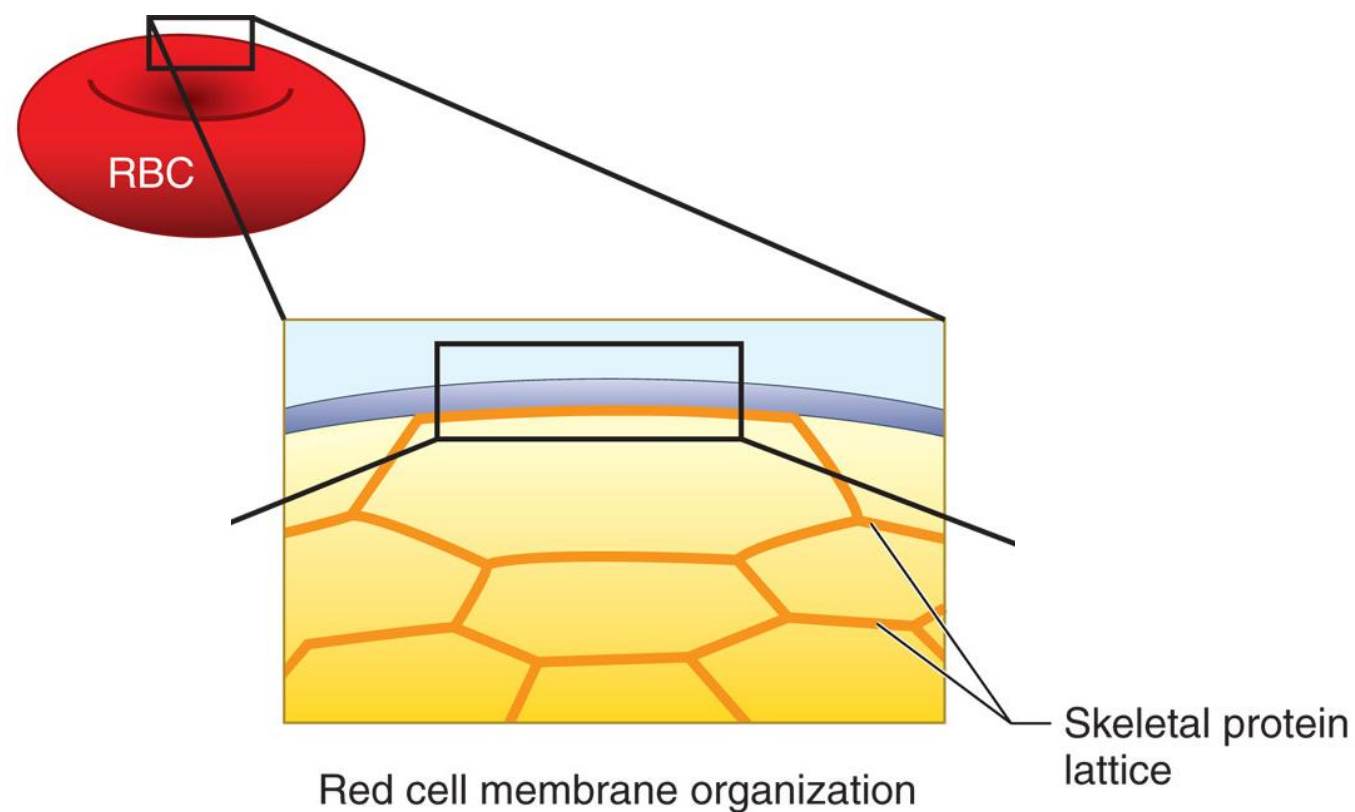


Lipid Composition

- ~95% of lipid content is equal amounts (1:1 occurrence)
 - Unesterfied cholesterol
 - Phospholipids (PLs)
- New lipids depend on exchange with plasma
- RBC shape change due to disruptions in distribution



Cytoskeleton

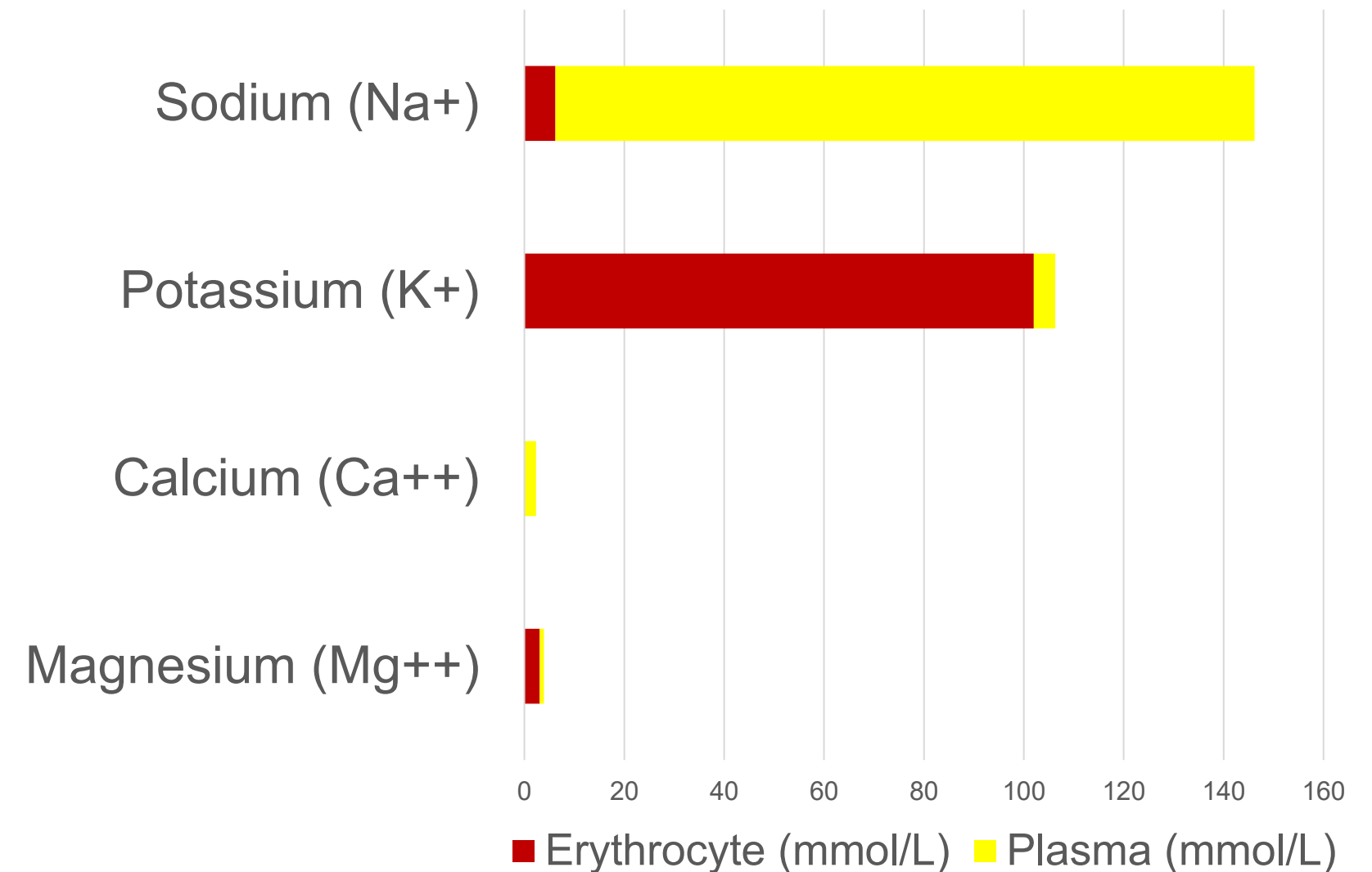


- Dynamic skeletal proteins
 - Continuous disassociation $\leftrightarrow$ association equilibrium
 - Allows deformability
- ~80% intracellular Ca^{++} found associated with RBC membrane
 - Low intracellular concentration : [extracellular]
 - [$\uparrow$] reduces deformability
- Cytoskeleton + Membrane allows shape rebound

Membrane Permeability

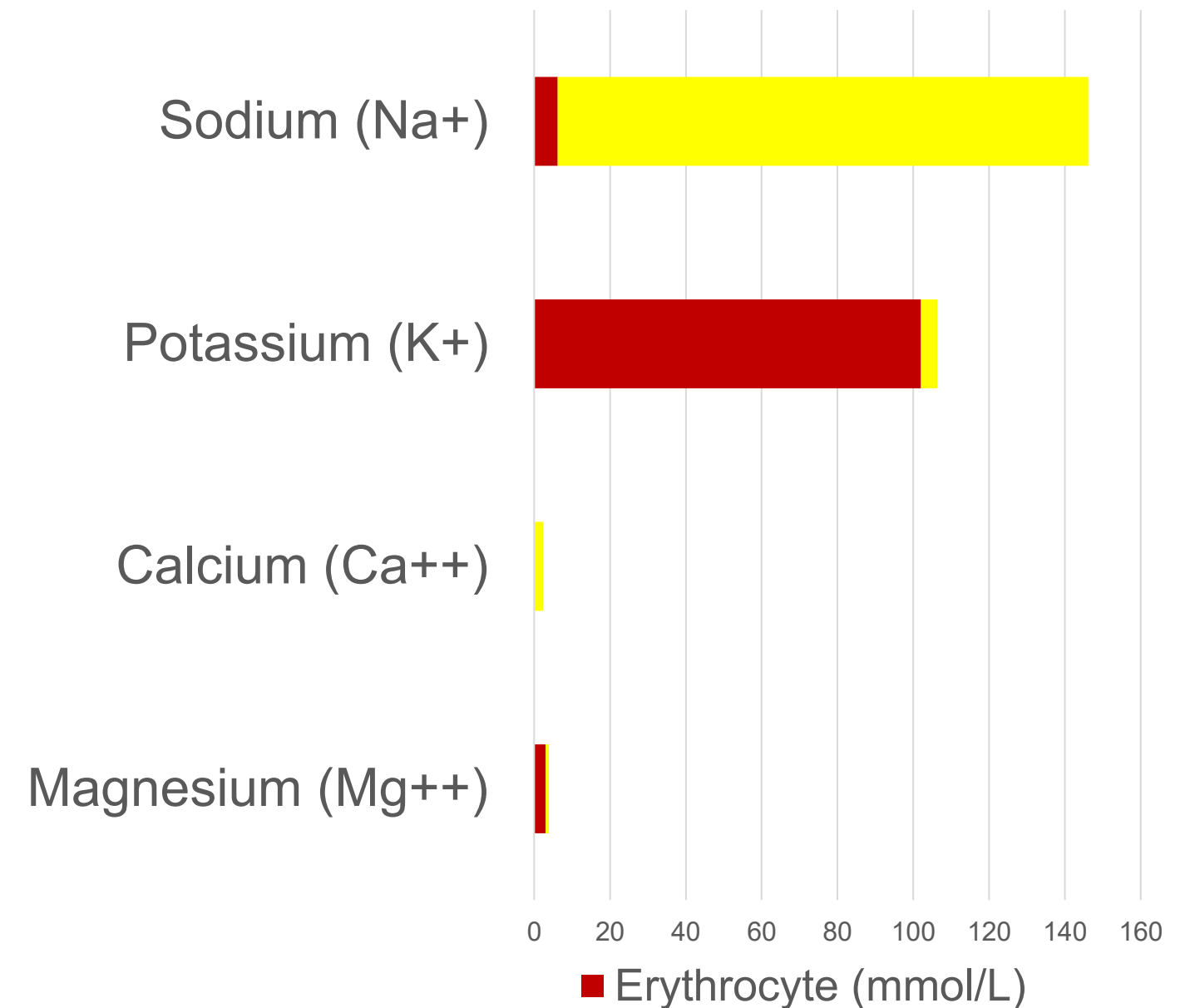
- Freely permeable to
 - Water, anions
- Nearly impermeable (ATP required)
 - Mono/divalent cations
 - Na^+ , K^+ / Ca^{++} , Mg^{++}
 - [Intracellular] $\neq$ [plasma]

Cation	Erythrocyte (mmol/L)	Plasma (mmol/L)
Sodium (Na^+)	5.4–7.0	135–145
Potassium (K^+)	98–106	3.6–5.0
Calcium (Ca^{++})	0.0059–0.019	2.1–2.6
Magnesium (Mg^{++})	3.06	0.65–1.05



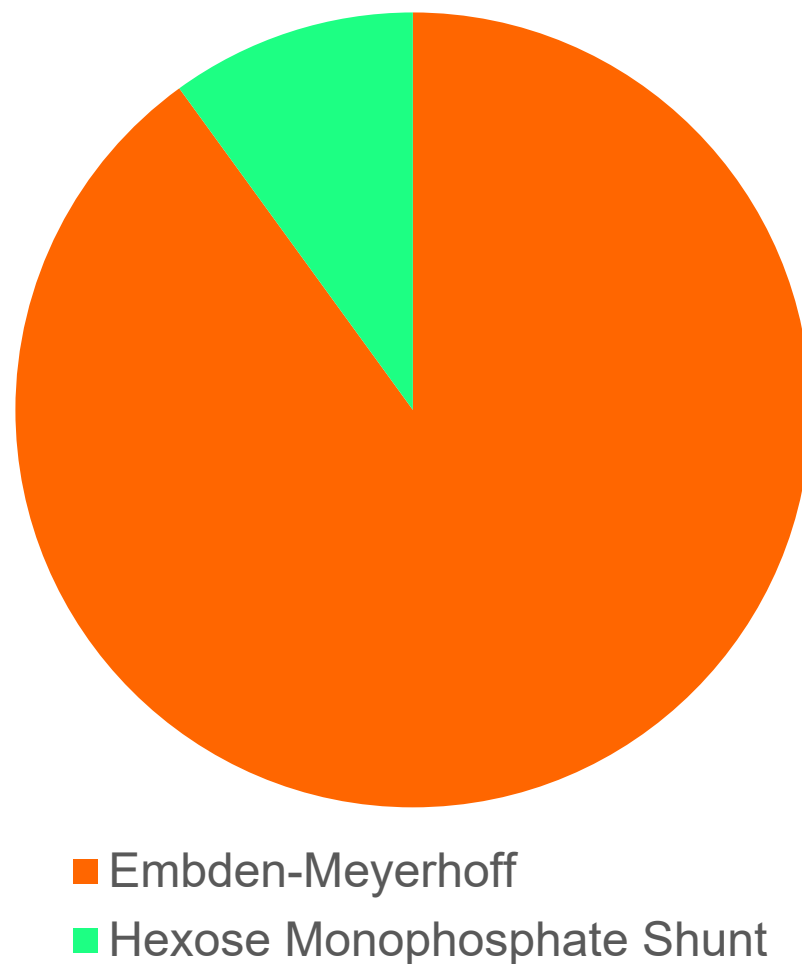
Membrane Permeability

- Osmotic equilibrium maintained by
 - Selective (low) permeability to cations
 - Cation pumps
 - Na^+/K^+ pump
 - 1 ATP = remove 3 Na^+ , uptake 2 K^+
 - $\uparrow \text{Ca}^{++}$ allows Na^+/K^+ movement along gradients
 - Ca^{++} pump
 - Maintains low intracellular concentration of Ca^{++}
- Glucose crosses via non-ATP transporter
 - Facilitated diffusion down its concentration gradient



Intracellular Metabolism

Anaerobic Glycolysis



- Anaerobic glycolysis (no mitochondria)
- ~90-95% glucose metabolized by glycolytic pathway
 - Embden-Meyerhof pathway
 - ATP
- ~5-10% glucose metabolized by
 - Hexose Monophosphate Shunt
 - $\text{HMP} + \text{G6PD} \rightarrow \text{NADPH}$
 - Maintains stability of hemoglobin
 - Disruptions = Heinz bodies (denatured hemoglobin)

Intracellular Metabolism

- Methemoglobin Reductase Pathway
 - O_2 dissociates from heme iron
 - Methemoglobin produced = iron in (ferric) state Fe^{+++}
 - Pathway + NADH maintains heme iron in reduced (ferrous) state Fe^{++}
- Rapoport-Leubering Shunt
 - Controls production of 2,3-BPG
 - Decreases O_2 affinity = releases O_2

Erythrocyte Destruction

- Mostly from senescence (age of RBC)
 - ~90% Extravascular
 - Within macrophages of spleen, bone marrow, liver
 - Recycles RBC components
 - ~10% Intravascular
 - Trauma during circulation results in lysis
 - Plasma proteins (haptoglobin, hemopexin) bind free hemoglobin
 - Transport to liver